I

(1)

Using equation (iv), under the steady-state condition, we have

$$\nabla \times \vec{B} = \mu_0 \vec{J}.$$

Taking the surface integral over an arbitrary surface S_C bounded by a closed curve C, we obtain

$$\int_{S_C} (\nabla \times \vec{B}) \cdot d\vec{S} = \mu_0 \int_{S_C} \vec{J} \cdot d\vec{S}.$$

Using Stokes' theorem,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_{S_C} \vec{J} \cdot d\vec{S},$$

which is exactly the relation described in the problem.

(2)

Consider a circular path $C(r)$ of radius r, centered on the z-axis and perpendicular to it.

Using the result derived in (1), we have

$$B_\phi(r)\, 2\pi r = \mu_0 I_{\text{enc}}(r).$$

For $r \leq R$, the current density is uniform:

$$\vec{J} = \frac{I}{\pi R^2} \hat{z}.$$

Thus, the enclosed current is

$$I_{\text{enc}}(r) = \frac{I}{\pi R^2} \cdot \pi r^2.$$

Therefore, we obtain

$$\boxed{B_\phi(r) = \frac{\mu_0 I r}{2\pi R^2}}.$$

(3)

For $r > R$, the enclosed current is I, so

$$B_\phi(r)\, 2\pi r = \mu_0 I.$$

Thus,

$$B_\phi(r) = \frac{\mu_0 I}{2\pi r}.$$

Combining the results,

$$\boxed{B_\phi(r) = \begin{cases} \dfrac{\mu_0 I r}{2\pi R^2}, & 0 \le r \le R, \\ \dfrac{\mu_0 I}{2\pi r}, & r > R. \end{cases}}$$

The magnetic field increases linearly for $r \le R$ and decreases as $1/r$ for $r > R$.

II

(4)

At the boundary $r = R$, the magnetic field is

$$B_\phi(R) = \frac{\mu_0 I}{2\pi R}.$$

The Lorentz force is given by

$$\vec{F}_B = q\vec{v} \times \vec{B}.$$

In our case, its magnitude is

$$F_B = e v_z B_\phi(R).$$

The instantaneous radius of curvature ρ satisfies

$$F_c = \frac{m v_z^2}{\rho}.$$

Since $F_c = F_B$, and the problem specifies $\rho = R/2$ at $r = R$, we have

$$\frac{m v_z^2}{R/2} = e v_z \frac{\mu_0 I_A}{2\pi R}.$$

Therefore,

$$\boxed{I_A = \frac{4\pi m v_z}{e\mu_0}}.$$

(5)

Substituting the given values into the result of (4), we obtain

$$|I_A| = \frac{4\pi(9.1 \times 10^{-31})(3.0 \times 10^8)}{(1.6 \times 10^{-19})(4\pi \times 10^{-7})}.$$

Thus, to two significant figures,

$$\boxed{|I_A| = 1.7 \times 10^4 \text{ A}}.$$

III

(6)

Using the vector identity

$$\nabla^2\vec{E} = \nabla(\nabla\cdot\vec{E}) - \nabla\times(\nabla\times\vec{E}),$$

together with Maxwell's equations, we obtain

$$\nabla(\nabla\cdot\vec{E}) - \nabla^2\vec{E} = -\mu_0\frac{\partial\vec{J}}{\partial t} - \mu_0\varepsilon_0\frac{\partial^2\vec{E}}{\partial t^2}.$$

Notice that

$$\nabla(\nabla\cdot\vec{E}) = \frac{1}{\varepsilon_0}\nabla\rho = 0,$$

since the charge density is spatially uniform within the beam.

Therefore,

$$\boxed{\frac{\partial^2\vec{E}}{\partial t^2} = c^2\nabla^2\vec{E} - \frac{1}{\varepsilon_0}\frac{\partial\vec{J}}{\partial t}}.$$

Hence,

$$\boxed{(A) = c^2\nabla^2\vec{E}, \qquad (B) = \frac{1}{\varepsilon_0}\frac{\partial\vec{J}}{\partial t}}.$$

IV

(7)

In the region $r > R$, there is no current density, so the wave equation becomes

$$\frac{\partial^2\widetilde{\vec{E}}}{\partial t^2} = c^2\nabla^2\widetilde{\vec{E}}.$$

Since the electric field has only a z-component,

$$\widetilde{\vec{E}}(r,t) = \widetilde{E}_z(r,t)\hat{z},$$

we only need to consider the z-component of the equation.

Thus,

$$\boxed{\left(\frac{\partial^2}{\partial t^2} - \frac{c^2}{r}\frac{\partial}{\partial r} - c^2\frac{\partial^2}{\partial r^2}\right)\widetilde{E}_z(r,t) = 0}.$$

(8)

Using

$$\widetilde{E}_z(r,t) = E_z(r)\cos(\omega t), \qquad k = \frac{\omega}{c},$$

we obtain

$$-\omega^2 E_z = c^2\left(\frac{1}{r}\frac{dE_z}{dr} + \frac{d^2E_z}{dr^2}\right).$$

Rearranging,

$$r^2\frac{d^2E_z}{dr^2} + r\frac{dE_z}{dr} + k^2r^2E_z = 0.$$

Let

$$x = kr, \qquad f(x) = E_z(r).$$

Then,

$$\frac{dE_z}{dr} = k\frac{df}{dx}, \qquad \frac{d^2E_z}{dr^2} = k^2\frac{d^2f}{dx^2}.$$

Substituting these expressions, we obtain

$$\boxed{x^2\frac{d^2f}{dx^2} + x\frac{df}{dx} + x^2f = 0},$$

which is the zeroth-order Bessel equation.

Therefore,

$$\boxed{(C) = kr}.$$

(9)

From Faraday's law,

$$\nabla \times \widetilde{\vec{E}} = -\frac{\partial \widetilde{\vec{B}}}{\partial t}.$$

The ϕ-component gives

$$\frac{\partial \widetilde{E}_r}{\partial z} - \frac{\partial \widetilde{E}_z}{\partial r} = -\frac{\partial \widetilde{B}_\phi}{\partial t}.$$

Since the electric field has only a z-component,

$$\frac{\partial \widetilde{E}_z}{\partial r} = \frac{\partial \widetilde{B}_\phi}{\partial t}.$$

Using

$$\widetilde{E}_z(r,t) = E_0 r^{-1/2}\cos\psi, \qquad \psi = kr - \omega t - \frac{\pi}{4},$$

we obtain

$$\frac{\partial \widetilde{B}_\phi}{\partial t} = -\frac{E_0}{2r^{3/2}}\cos\psi - \frac{kE_0}{r^{1/2}}\sin\psi.$$

Integrating with respect to time,

$$\widetilde{B}_\phi(r,t) = \frac{E_0}{2\omega r^{3/2}}\sin\psi - \frac{kE_0}{\omega r^{1/2}}\cos\psi + C(r).$$

Since

$$\left\langle \widetilde{B}_\phi(r,t)\right\rangle_T = 0,$$

we have $C(r) = 0$.

Therefore,

$$\boxed{\widetilde{B}_\phi(r,t) = \frac{E_0}{2\omega r^{3/2}}\sin\psi - \frac{kE_0}{\omega r^{1/2}}\cos\psi}.$$

(10)

The Poynting vector is

$$\vec{S} = \frac{1}{\mu_0}\widetilde{\vec{E}} \times \widetilde{\vec{B}}.$$

Its radial component is

$$S_r = \frac{1}{\mu_0}\left(\widetilde{E}_\phi \widetilde{B}_z - \widetilde{E}_z \widetilde{B}_\phi\right).$$

Since the fields have only E_z and B_ϕ components,

$$S_r = -\frac{1}{\mu_0}\widetilde{E}_z \widetilde{B}_\phi.$$

Substituting the results from (9),

$$S_r = -\frac{E_0^2}{2\mu_0\omega r^2}\cos\psi\sin\psi + \frac{kE_0^2}{\mu_0\omega r}\cos^2\psi.$$

Taking the time average over one period,

$$\langle\cos\psi\sin\psi\rangle_T = 0, \qquad \langle\cos^2\psi\rangle_T = \frac{1}{2}.$$

Therefore,

$$\boxed{\langle S_r(r)\rangle_T = \frac{kE_0^2}{2\mu_0\omega r}}.$$

(11)

As $x \to \infty$, if we neglect the middle term of equation (vi), we obtain

$$x^2\frac{d^2f}{dx^2} + x^2 f = 0.$$

Thus,

$$\frac{d^2f}{dx^2} + f = 0,$$

whose solution is

$$f(x) = E_0 \cos(x + \delta).$$

For an outward-propagating wave, this corresponds to an electric field of the form

$$\widetilde{E}_z(r, t) = E_0 \cos(kr - \omega t + \delta).$$

The amplitude is independent of r. Consequently, the time-averaged radial Poynting flux is also independent of r:

$$\langle S_r(r) \rangle_T \propto r^0.$$

Consider a cylindrical surface of unit length along the z-axis. The total energy flux through this surface is proportional to

$$P(r) = 2\pi r \langle S_r(r) \rangle_T.$$

Therefore,

$$P(r) \propto r.$$

This means that the total outward energy flux increases with r, which violates energy conservation.

Hence, neglecting the middle term does not give a physically acceptable solution.